Roll No: _____

Amrita Vishwa Vidyapeetham, Amritapuri Campus
Midterm Examinations, October 2025

Integrated MSc/ MSc (Hons) Physics and Mathematics
Third Semester

24PHY206 Waves

Duration: 2:00 hours | Maximum : 50 Marks

***Answer all questions**. Perform all calculations up to 3 significant digits. Calculators are permitted, but **exchange of calculators is not permitted**.*

1. (a) Derive the differential equation governing the motion of a simple harmonic oscillator and obtain the corresponding expression for its displacement as a function of time.
[CO1,L2][7 marks]

   (b) A particle executing simple harmonic motion in a straight line has velocities 10cm/s and 12cm/s as it passes through points 8cm and 5cm respectively from the equilibrium position. Find the amplitude of oscillation. [CO1, L3][3 marks]

2. (a) Explain the working of an LCR circuit as an example of a forced harmonic oscillator and derive an expression for its impedance. [CO1, L2][6 marks]

   (b) Given that for an LCR circuit , L= 50mH,C=100μF, R= 10Ω for Vrms= 15V. Find the Q-value of the circuit and voltage across the inductor at resonance. [CO1, L3][4 marks]

3. Describe the properties of modes of a system with two degrees of freedom, in terms of frequency, phase constants, occurrence of equilibrium positions. [CO2, L2][3 marks]

4. A coupled pendulum with length $l = 1\text{m}$, $M = 0.2$ kg, coupled by a spring of spring constant $K = 0.4\ \text{N/m}$, $g = 9.8\ \text{m/s}^2$. The normal mode frequencies are [CO2, L3][7 marks]

$$\omega_1^2 = \frac{g}{l}, \qquad \omega_2^2 = \frac{g}{l} + \frac{2K}{M}.$$

   (a) Find the normal mode angular frequencies, the average frequency ω_{av}, modulation frequency ω_{mod}, beats frequency ω_{beats}. [2.5]

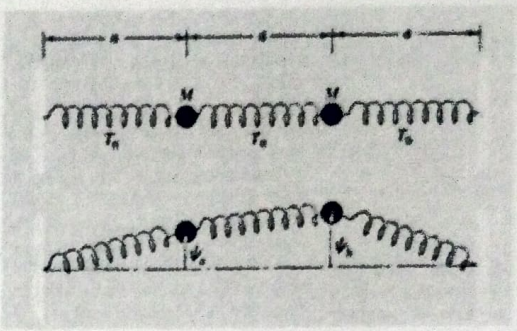
   (b) Write down the displacement $\psi(t) = \psi_1 + \psi_2$ in terms of modulation amplitude, average and modulation frequencies, ω_{av}, and ω_{mod} (don't derive); plot the function. [2]

   (c) What is the frequency with which energy transfer takes place between the two masses? Does does it differ from that of the variation of amplitudes of the masses? Explain with a plot and in a couple lines. [2.5]

5. Consider the transverse oscillations of two coupled masses. The equations of motions read

$$M\frac{d^2\psi_a}{dt^2} = -\frac{T_0}{a}\psi_a + \frac{T_0}{a}(\psi_b - \psi_a),$$

$$M\frac{d^2\psi_b}{dt^2} = -\frac{T_0}{a}\psi_b - \frac{T_0}{a}(\psi_b - \psi_a).$$



Please refere to figure on the right. Symbols carry usual meanings.

(a) Looking for displacement solution of the form

$$\psi_a = A\cos(\omega t + \varphi), \quad \psi_b = B\cos(\omega t + \varphi),$$

derive the normal mode angular frequencies ω_1 and ω_2 and the ratios of amplitude for each mode using the systematic method. [CO2, L3][8 marks]

(b) Sketch both modes, and write down the general solution for the displacements. [CO2, L3][2 marks]

6. In a Melde's experiement, transverse oscillations excited in a continuous string of length $L = 1.25$ m, linear densit $\rho_0(\text{linear}) = 1.2$ g/m is kept under a tension with mass $M = 100$g, $T = 0.98$ N. Dispersion relation is $\omega = v_0 k$.

(a) Briefly explain and establish the connectin between the quantized frequencies of normal modes of transverse oscillations of a string of length L fixed at both ends and the boundary condition of the ampitude, $A(z) = A\sin kz$, at $z = L$. [CO3, L3][3 marks]

(b) Find (i) speed of transverse waves in the string, (ii) the wavelength and frequency of fundamental harmonic. [CO3, L3][3 marks]

7. Consider now the transverse waves in a continuous string. Use the same equilibrium tension $T_0 = 0.98$ N, and the linear density $\rho_0(\text{linear}) = 1.2$ g/m. Taking the frequency of the mechanical driver $\nu = 50$ Hz, and amplitude of the wave $A = 3$cm, find [CO3,L3][4 marks]

(i) the phase velocity,
(ii) the characterstic impedance, and
the average power transmitted P_{av}. How is it related to the average energy density?